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Insights into genetic architecture and disease associations of genes associated with different human blood group systems using codon usage bias

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ABSTRACT

The differential use of synonymous codons of an amino acid is an imperative evolutionary phenomenon, termed codon usage bias, that functions across various levels of organisms. It is accustomed to providing an understanding of a gene's differential architecture driven by functional regulation of gene expression. Numerous synonymous mutations are linked to various diseases, demonstrating that silent mutations can be deleterious. We employed bioinformatics methods to examine codon usage trends in 263 coding sequences of 44 blood group systems. The blood group systems were categorized into two groups based on association with a sort of neurodegenerative disorder. We performed a CUB study to investigate how multiple components, such as selection, mutation and biased nucleotide composition are accountable for the evolution of the transcripts of the blood group antigens. The compositional analysis implicated blood group genes were GC-rich. RSCU analysis showed G/C-ending codon choice among synonymous codons. Also, a distinct codon choice was found in both blood groups for serine and proline. Moreover, the leucine-coding CTG codon was found the most overrepresented in all the genes, indicating selectional pressure substantially impacts overall codon usage. This was also supported by biplot analysis. Additionally, CpC and GpG overrepresentation is in concordance with the results concerning neurodegenerative disorders where CpC has been attributed to non-CpG methylation and linked to several neurodegenerative ailments. Both the Z-test analysis and rare codon choice showed a substantial difference in codon usage by the genes of both groups.

Abbreviations: AT%: total A and T nucleotides available; AT3%: nucleotides A and T present at the third position of codon; CAI: codon adaptation index; CDC: codon deviation coefficient; CDSs: coding sequences; CpC: cytosine per cytosine; CpG: cytosine per guanine; CUB: codon usage bias; GC%: total G and C nucleotides percent; GC12: the average of nucleotide G and C present at the first and second position of codons; GC3: total G and C nucleotides present at the third position; ENc: effective number of codons; FASTA: fast alignment; NCBI: National Center for Biotechnology Information; P2: translational selection; RSCU: relative synonymous codon usage; TpG: thymine per guanine

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Introduction

The blood group systems are incredibly polymorphic, with the International Society of Blood Transfusion listing over 45 different blood group systems encompassing over 362 antigens as of November 2023 (*Red Cell Immunogenetics and Blood Group Terminology | ISBT Working Party | The International Society of Blood Transfusion* (ISBT)). Our knowledge of blood groups has grown through time to include particular disease associations with red blood cell (RBCs) surface antigens. Surface markers, or antigens, are found on the cells that compose the tissues and organs of the body. The same is true for RBCs. An individual's blood group is decided by the antigens expressed on their RBC. These antigens are either sugars (ABO blood group) or proteins (Rh blood group) that are synthesized using a series of reactions including the migration of sugar units, which are then processed by enzymes. A person's DNA determines the kind of enzymes they have and, consequently, the kind of sugar/protein antigens that build up on their RBCs. Thus, a blood group system is genetically unique and consists of antigens that are governed by one or more closely interconnected loci (Dean, 2005). The functional categorization of human blood group antigens, the enzymes that synthesize them, and their association with disease risks have all been elucidated owing to recent developments in technology, biochemistry and genetics.

ABO blood group antigens and Rh blood group systems have strong expression of a majority of human cells and tissues and have both been linked in the pathogenesis of a variety of systemic disorders such as hemolytic disease of

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the newborn, cancer, vascular disorders (cerebral arterial ischemia, venous thromboembolism, and angina), cognitive impairment and metabolic diseases (hypertension, dyslipidemia, obesity, diabetes and cardiovascular disease) (Ewald & Sumner, 2016). ABO polymorphisms can impact the pathophysiology of cells and individuals (Yamamoto et al., 2012). A prior study analyzing 1,217 disease categories in 5.1 million Swedish citizens showed that the majority of populations had no apparent relationship to the RhD or ABO blood classification systems. Despite this, 49 ailments were shown to be associated with ABO blood types and one with RhD status, according to the analysis. A noteworthy discovery was that kidney stones are less common in those with blood group B (Dahlén et al., 2021). For these consequences, research on blood group systems cannot merely be confined to transfusion and transplantation medicine alone.

Furthermore, different studies focusing on the relationship between the blood group system and aging-related neurological ailments were conducted, but the findings were inconsistent. In a study, on comparing to controls, the ABO blood group distribution in 187 Parkinson's disease patients in Northern Ireland indicated a deficit in blood group A and an excessive amount of blood group B. The relative incidence of blood group B in contrast to group A was found significantly greater than unity ($p < 0.05$) (Kak et al., 1970). In contrast, two investigations conducted by Strang (Strang, 1965) and Chia (Chia & Liu, 1992) reported no significant variations in ABO blood type dissemination in Parkinson's disease patients when weighed against controls. Another case-control study involving 495 cases with cognitive impairment showed the possibility of poorer cognitive trajectories with non-O blood type adults. The study discovered that having the blood group AB with high levels of clotting factor VIII was associated with an increased risk of cognitive impairments (Alexander et al., 2014). A recent study based on volumetric variations in grey matter between O (76 cases) and non-O (113 cases) normal people, discovered that O blood type was correlated with greater amounts of grey matter in the cerebellum and temporal-mediotemporal/limbic regions than non-O blood type, implying that O blood type may protect against age-related neurodegenerative conditions, particularly Alzheimer's disease, that are linked to volumetric loss in these cerebral areas (De Marco & Venneri, 2015). Another study indicated a positive correlation with elevated levels of certain hemostatic markers, such as von Willebrand factor, in vascular dementia and cognitive impairment (Quinn et al., 2011). Factor VIII and von Willebrand factor (vWF) plasma levels are mostly controlled by the ABO blood group. Individuals with blood group O exhibit approximately 25% lesser plasma levels of these glycoproteins than those with blood group A (Yamamoto et al., 2012). Such blood group systems linked to neurodegenerative disorders must be envisaged to investigate the molecular characteristics of the subgroup of genes implicated.

Disease heritability is influenced significantly by rare and low-frequency variations. Because rare variations are frequently spurious, running an association test on a single locus can be challenging because the test will be underpowered (Lee et al., 2014). To combat this problem, rare variants can be grouped based on prior biological knowledge about the disease's genetic etiology. Filtering for a specific type of variant, such as non-synonymous changes, has significant advantages in focusing the association on variants that are more influential in determining the phenotype and making the results easier to interpret (Kim et al., 2017).

Synonymous mutations occur when the coding sequence of a gene is altered at the nucleotide level despite affecting the amino acid sequence. These features of the genome are likely to be part of what leads researchers to believe that synonymous mutations and variations have little to no influence on the protein. Nevertheless, it has been demonstrated that different organisms prefer different codons over others, and this adoption can also vary between genes within the same organism, implying that synonymous codons have undergone evolutionary pressure to improve (Miller et al., 2018). Furthermore, synonymous variants have been linked to a variety of disorders, including neurological, immunological, cancer, cardiovascular and other diseases (Sauna & Kimchi-Sarfaty, 2011). Since several synonymous mutations are linked to a variety of diseases, demonstrating that silent mutations can be deleterious, it will be crucial to examine several types of codon bias.

The use of computational approaches to predict gene expression levels is appealing since it eliminates the need for expensive and complex experiments. Highly expressed genes in a variety of organisms have been anticipated by recent research, offering substantial novel perspectives on gene expression (Karlin et al., 2004; Merkl, 2003; Roymondal et al., 2009; Wagner, 2000). These highly expressed genes frequently encode proteins that are critical to cell physiology, including, transcription/translational factors, and those involved in biosynthesis, degradation, and metabolism. Such genes are biologically significant as their biased amino acid composition may reduce the energy expenses associated with biosynthesis. Further evidence suggests that codon bias directs gene expression through translational rate optimization, allowing highly expressed genes to be distinguished from average genes based on biased codon usage. To quantify the extent of codon bias, several research presented different codon utilization patterns and indicators (Parvathy et al., 2022; Sharp & Li, 1987; Wright, 1990).

A phenomenon known as codon usage bias is caused by the adoption of synonymous codons at variable frequencies, according to numerous earlier research. The evolution of CUB is a complicated and contentious topic. The neutral theory and the selection-mutation drift balance model theory are two predominantly recommended theories for explaining the development of synonymous codon usage alteration, or CUB. The impact of these evolutionary factors on diverse species, however, is uncertain. In addition, biological parameters including GC content, degree of gene expression, gene length, protein organization and tRNA quantity, have all been linked to CUB (Chakraborty et al., 2019).

Codon usage can be skewed in a variety of ways. The relative synonymous codon usage (RSCU) score indicates how frequently a codon is utilized in comparison to other

synonymous codons, providing a measure to assess whether a mutation substitutes a more frequent codon with a rarer codon or vice versa (Sauna & Kimchi-Sarfaty, 2011). *In vitro*, replacing rare and frequent synonymous codons may have an impact on translation and protein function (Komar et al., 1999). Codons that use rare and common tRNAs at the start and end of the gene, respectively, are used to govern the translation process (Tuller et al., 2010). Furthermore, codons that are better optimal than others, have stronger connections with their cognate tRNA or more tRNAs available, allowing less pausing and higher fidelity translation, and in some situations, impacting mRNA stability (Presnyak et al., 2015).

We performed CUB study to investigate how multiple components, such as selection, mutation and composition are accountable for the evolution of the coding sequences of the blood group antigens that have an association with some kind of neurodegenerative disorders as no previous research has been reported. In this study, we used NCBI to retrieve available coding sequences of 13 blood group systems (ABCC1, ABO, Cartwright, Chido, CTL2, CD36, Kell, Knops, Kx, Kanno, Lewis, John Milton Hagen, and Xg) out of a total of 44 that are associated with neurodegenerative disorders (Table 1). We have done a comprehensive study employing several CUB indices to investigate unusual codon utilization, codon usage bias, and their underlying causes. We analyzed compositional attributes, dinucleotide contents, RSCU, parity and neutrality plot, principle component analysis and discovered variations in the synonymous codon usage among those blood group systems that also have an association with neurodegenerative disorders inheritance. Relationships between disease incidence and blood types may eventually open up new study directions and aid in our understanding of the underlying molecular mechanisms of diseases.

In order to improve clinical and biological understanding, translational bioinformatics research combines molecular data (such as DNA, RNA, proteins, small molecules and lipids) with clinical data (such as patients, diseases, symptoms, pathology reports, clinical imaging and medications). It helps in cohort finding and correlating clinical data with genomic and cellular data leading to novel discovery. Recent advances in databases have boosted this domain. High-throughput biological assays for full genome, transcriptome, proteome, and metabolome assessment; access to substantial clinical information *via* digitized clinical records; and advancements in computers for storing and maintaining information, and visual representation are key contributors to translational bioinformatics (Altman, 2012). Additionally, molecular networks, like those proposed by Iossifov et al., are useful for interpreting genetic data and have shown potential in linking molecular interactions with genetic models for disorders like autism, bipolar disorder, and schizophrenia, supporting the idea that these ailments may exist on a spectrum rather than as distinct conditions (Iossifov et al., 2008). This technique can be further utilized to predict unknown blood group system genes that might be associated with disease incidence. It can also help clinicians better understand the

Table 1. List of blood group systems that are associated with neurodegenerative disorder(s).

S.No.	Blood group systems	ISBT code	Symbol	Gene	Linked disorders/relevance to neurodegeneration	References
1.	Cartwright/Yt	011	YT	ACHE (Acetylcholinesterase)	Parkinson's disease (PD), Olivopontocerebellar Atrophy, Spinocerebellar Ataxia and Progressive Supranuclear Palsy	(Rao et al., 1993; Walczak-Nowicka & Herbet, 2021)
2.	Knops	022	KN	CR1	Alzheimer's disease	(Daniels et al., 2004; Shen et al., 2015)
3.	Xg	012	XG	XG, CD99	Dementia	(Manuello et al., 2024)
4.	Kell	006	KEL	KEL	McLeod Syndrome	(Redman & Lee, 2013)
5.	Kx	019	XK	XK	McLeod Syndrome	(Daniels, 2004)
6.	Chido/ Rodger	017	CH/RG	C4A, C4B	Systemic lupus erythematosus.	(Lokki et al., 1999; Mougey, 2019; Wilson & Perez, 1988; Y. Yang et al., 2007)
7.	Lewis	007	LE	FUT3	Fucosidosis type I causes severe neurodegeneration	(Jajosky et al., 2023; Stepien et al., 2020)
8.	John Milton Hagen	026	JMH	SEMA7A	Parkinson's disease, multiple sclerosis	(Quintremil et al., 2018)
9.	ABO	001	ABO	ABO	Alzheimer's disease, cognitive impairment	(Alexander et al., 2014; De Marco & Venneri, 2015)
10.	Kanno	037	KANNO	PRNP	Creutzfeldt-Jakob disease, fatal familial insomnia, Gerstmann-Straussler disease, Huntington disease-like 1, and Kuru	(Baldwin & Correll, 2019; Matsubayashi & Sanjo, 2022)
11.	CD36	045	CD36	CD36	Alzheimer's disease, Parkinson's disease, potential involvement in neuroinflammation, neurovascular dysfunction, and subsequent neurodegeneration.	(Abumrad & Moore, 2011; Dobri et al., 2021; loghen et al., 2021)
12.	ABCC1 (Krohn et al., 2011; Puris et al., 2022)	043		ABCC1	ABCC1	Alzheimer's disease
13.	CTL2	039	CTL2	SLC44A2	Autism disorder, Huntington's disease	(Vignone et al., 2022)

relationship between different geographical populations and blood group system genes without the need for expensive and complex experiments.

Methodology

Data retrieval

The nucleotide sequences for different Blood group antigen coding genes in *Homo sapiens* were retrieved from National Center for Biotechnology Information (NCBI) https://www.ncbi.nlm.nih.gov/Genebank in FASTA format. The coding sequences (CDSs) of 44 different Blood group antigen coding genes were used in this study while CDSs having less than 250 base pair counts were exempted. A total number of 263 CDSs which correspond to 128787 codons were investigated in this study (Supplementary Table 1), where 64 CDSs are associated with neurodegenerative disorders. All 44 blood group systems were categorized into two groups for the comparative study; group 1 consisted of neurodegenerative disorder associated 13 blood group systems (Supplementary Table 2) and group 2 consisted of non-neurodegenerative disorder associated 31 blood group systems (Supplementary Table 3).

Nucleotide content analysis

In nucleotide content analysis, specifically at the third nucleotide codon position frequency i.e., T3%, G3%, C3% and A3% were evaluated. Using the CAI calculator (http://genomes.urv.es/CAIcal/) the overall GC%, GC12 and GC3, and overall AT%, AT3% were analyzed (Puigbò et al., 2008).

Relative synonymous codon usage (RSCU) analysis

RSCU is the ratio of an amino acid's observed frequency to its predicted frequency provided all synonymous codons have been allocated uniformly (Sharp & Li, 1986). The RSCU value for 263 CDSs was analyzed in this study from the CAIcal Server (http://genomes.urv.es/CAIcal/); the value less than 0.6 denotes codons corresponding to them are underrepresented and values more than 1.6 are overrepresented, values between them considered to be unbiased (Sharp & Li, 1987).

Relative codon adaptation (RCA)

To utilize the potential of relative codon frequencies as an indicator of gene expression without requiring genomic corrections or limitations to gene length, RCA is applied to any reference sequence set. The RCA index first determines the expected frequency of a codon, considering its positional base frequencies for an individual reference set. The difference between the observed and expected codon frequencies is then used to determine the codon adaptation. Therefore, when calculating each RCAxyz term (codon xyz in the CDS of interest), RCA explicitly considers sequence composition, and as a result, it should be able to deliver more reliable and precise gene expression estimations compared to other codon bias indices (Fox & Erill, 2010). The RCA value for 263 CDSs of different blood group systems was analyzed in this study using the GenRCA Server (https://www.genscript.com/tools/rare-codon-analysis).

Relative dinucleotide abundance analysis

Dinucleotide relative abundance or odds ratio is the corresponding proportion of the observed to the predicted dinucleotide frequency—particularly for CG (CpG) and TA (TpA); if this ratio is below 0.78 the dinucleotide is underrepresented while the value more than 1.25 the dinucleotide is overrepresented (Kunec & Osterrieder, 2016). In the current study dinucleotide relative abundance (odds ratio) was determined using the Genomatix software suite (http://emboss.bioinformatics.nl/cgi-bin/emboss/compseq).

Rare codon analysis

Even though optimal codons are commonly considered a key factor in enhanced gene expression, the importance of rare codons should not be overlooked. The ramp sequence, a 20–40 nucleotide stretch at the gene's 5′ end that governs the translation rate and, consequently, influences appropriate protein folding, might include rare codons (Liu et al., 2021). In addition, rare codon inclusion may alter the translation rate, affecting the yield and solubility of proteins (Rosano & Ceccarelli, 2009). Anaconda 2.0 software was deployed to analyze rare codons (Moura et al., 2007). Codons with occurrence frequencies of less than 0.5% were treated as rare.

Analysis of the effective number of codons (ENc) and effective number of codons prime (ENc′)

ENc and its variant ENc′ are done to identify the bias in the use of synonymous codons. ENc′ is an improved version of ENc that describes how the observed codon usage deviates from an expected distribution and depends on Pearson's X2 statistics (Satapathy et al., 2017). Most amino acids have two or more codons corresponding to them known as synonymous codons. ENc values range between 20 and 61 where the lowest ENc score depicts the amino acids are encoded by a particular codon only, indicating extreme bias while the value 61 implicates no propensity in available codons utilization, depicting codons corresponding to the amino acid are used equally (Wright, 1990). In this study, the ENc was computed using the CAIcal Server (http://genomes.urv.es/CAIcal/), and ENc′ by using an open-source platform available at (http://agnigarh.tezu.ernet.in/~ssankar/cub.php) (Satapathy et al., 2017).

Codon adaptation index (CAI) analysis

CAI value shows the favored codon used among highly expressed genes. CAI is used to assess the expression of the gene solely based on its codon composition. The CAI value of all 263 transcripts was calculated using the CAIcal Server

(http://genomes.urv.es/CAIcal/). CAI value ranges from 0 to 1, the higher the value of CAI the greater the chances of gene expression (Sharp & Li, 1987). *Homo sapiens* codon usage table available at Kazusa Codon Usage Database (https://www.kazusa.or.jp/codon/) was used as the reference set for CAI calculation.

Codon deviation coefficient (CDC) analysis

For evaluating the degree and patterns of CUB across genes or genomes possessing diverse sequence compositions, the CDC analysis offers an extensive measure of codon usage bias. The CDC analysis does not require a prior understanding of reference gene sets because it quantifies CUB using the cosine distance metric and uses bootstrapping to establish its relevance. This method determines CUB by considering background nucleotide compositions (the GC content of the coding sequence) customized to codon positions. The range of CDC values is 0 to 1, where a number closer to 0 suggests a lower amount of CUB and a closer match between the observed and expected patterns of codon usage. A value nearby to 1 implies a greater degree of codon usage bias (Z. Zhang et al., 2012). The CDC value for 263 CDSs of different blood group system genes was computed in this study using the GenRCA Server (https://www.genscript.com/tools/rare-codon-analysis) (Fan et al., 2024).

Neutrality plot analysis

A neutrality plot between GC12 vs GC3 is employed to analyze the effect of mutational selection and translation selection in determining the CUB. In the neutrality plot, GC12 is ordinate and GC3 is abscissa, the regression line's gradient represents the pressure for mutation pressure (N. Kumar et al., 2016).

Parity plot analysis (PR2)

PR2 bias was computed by employing AT bias [A3/(A3 + T3)] as the ordinate and GC bias [G3/(G3 + C3)] as the abscissa (Chen et al., 2014; Sueoka, 1988). If the implications of natural selection and mutation pressure have been equalized, the coordinates will lie in the middle of the plot (0.5, 0.5), where A = T and G = C. The degree of divergence from the center indicates bias (Wu et al., 2015).

ENc-GC3 plot analysis

An ENc-GC3 plot is plotted between the ENc values as ordinate and GC3 values as abscissa. An ENc-plot is generally employed to determine mutation bias that influences the codon utilization pattern. The plot's ENc values indicate the extent to which natural selection and other factors influence codon usage bias. Conversely, ENc values that are nearer to the standard curve reflect the significant influence of mutational pressure on CUB (Wright, 1990).

Translational selection (P2)

The P2 value greater than 0.5 suggests a bias concerning translational selection (Gouy & Gautier, 1982). The following formula is used for calculating the P2 value:

$$P2 = \frac{(WWC + SSU)}{(WWY + SSY)}$$

Here, W = A or T, S = C or G, and Y = C or T P2 value helps in the assessment of codon-anticodon interaction and translation efficiency in a gene (Bennetzen & Hall, 1982). The P2 value for 263 CDSs of different blood group system genes was computed in this study using the GenRCA Server (https://www.genscript.com/tools/rare-codon-analysis) (Fan et al., 2024).

Principal component analysis (PCA)

A multivariate statistical method widely used in codon usage analysis, where axis 1 is plotted against axis 2 resulting in an analysis of the prevailing tendency of codon usage patterns within the different blood groups (Chen et al., 2017). In this study, the average RSCU values of each blood group are studied, the 59 dimensional vectors each representing the 59 synonymous codons excepting AUG (Met), UGG (Trp), and three termination codons UAA, UAG and UGA. The PCA analysis was done by PAST4.03 software.

Statistical analysis

Correlation between overall nucleotide composition and its composition at the third codon position, different plot analyses such as neutrality plot, parity plot, and ENc-GC3 plot including all the statistical analyses were performed using the SPSS 18 software. The PCA analysis and Z-test of RSCU values were done by PAST4.03 software.

Results and discussion

G/C-ending codons preference in the compositional analysis showed implications for gene expression and codon utilization patterns

One of the main elements thought to influence how genes and genomes use codons is the makeup of nucleotides. We primarily investigated the nucleotide composition with GC-content representing the general trend of codon mutation (Jernigan & Baran, 2002). The overall average values of Group 1 (neurodegenerative associated blood group systems) reveal that T % (23.04 ± 4.36) was the lowest, while C % (27.06 ± 5.18) was the highest, while A% and T% were nearly equal with values of 23.44 ± 4.87 and 23.04 ± 4.36, respectively (Supplementary Table 2). The nucleotides in the third position play an important role in coding protein sequences, with A3% (16.27 ± 8.16) being the lowest, followed by T3% (20.82 ± 9.55), G3% (28.95 ± 8.77), and C3% (33.94 ± 9.26) being the highest. This study also discovered that all of the genes are GC rich, as the GC % (53.51 ± 8.87) and GC3% (62.90 ± 17.44) were higher than the AT % (46.48 ± 8.87) and

Table 2. Average values of gene length, nucleotide contents (%A, %C, %T, %G, %AT, %GC, %A3, %C3, %T3, %G3, %AT3, %GC3, %GT, and %AC), ENc, ENc′, CDC, CAI, RCA, and P2 of all 263 envisaged blood group system genes.

S.No.	Compositional indices	Overall average ± SD
1.	Gene length	1471.027 ± 1158.175
2.	%A	23.343 ± 4.575
3.	%C	26.082 ± 4.478
4.	%T	25.095 ± 4.338
5.	%G	25.481 ± 3.925
6.	%G + C	51.562 ± 7.926
7.	%A + T	48.438 ± 7.926
8.	%A3	16.800 ± 7.227
9.	%C3	32.641 ± 8.483
10.	%T3	22.049 ± 7.568
11.	%G3	28.510 ± 6.897
12.	%G3 + C3	61.152 ± 14.347
13.	%A3 + T3	38.848 ± 14.347
14.	%G + T	50.575 ± 2.557
15.	%A + C	49.425 ± 2.557
16.	Enc	50.262 ± 6.868
17.	ENc′	50.286 ± 3.072
18.	CDC	0.117 ± 0.039
19.	CAI	0.761 ± 0.048
20.	RCA	1.099 ± 0.027
21.	P2	0.501 ± 0.060

AT3% (37.10 ± 17.44), respectively. Additionally, in the Group 2 blood group systems (Supplementary Table 3) the average nucleotides composition revealed that A % (23.30 ± 4.47) was the lowest, while C % (25.76 ± 4.17) was the highest, similarly the nucleotides at the third position were observed in the following order i.e., A3% (16.96 ± 6.89) being the lowest, followed by T3% (22.44 ± 6.75), G3% (28.36 ± 6.16), and C3% (32.22 ± 8.17) being the highest. Furthermore, GC3% (60.58 ± 13.14) was found higher as compared to AT3% (39.41 ± 13.14). Based on these findings, the overall composition of nucleotides suggests that compositional constraints may influence the pattern of codon utilization in both Groups' genes. An incredibly low or extremely high GC composition typically corresponds to a more substantial CUB. Thus, codon usage bias and GC composition are correlated (Wan et al., 2004).

However, in the Knops blood group system genes, AT% (50.69 ± 0.067) preferred over GC% (49.31 ± 0.067), likewise, in the third position AT3% (52.94 ± 0.13) preferred over GC3% (47.05 ± 0.13) indicating AT richness. A similar trend in compositional preference of the CD36 blood group system gene was found. Except for the Knops and CD36 blood group systems, the blood group genes of Group 1 were primarily found to be GC-rich. GC content at the third codon position indicates CUB. According to Shen et al. (2015) (Shen et al., 2015), genes that have abundant GC3 content co-translate in particular spatial locations and possibly play a role in the architecture of the genome (Kumar et al., 2021). Additionally, the ability of the GC content to retain the coding region of DNA in an open chromatin state implicates it in the active transcription process as well. Based on prior studies, highly expressed genes might possess decreased mutation rates, owing to DNA repair processes (Hoeijmakers, 2001). In general, synonymous codons differ at the third codon position, and the degree of synonymous CUB can be precisely determined by looking at the GC3 content (Shen et al., 2015; Uddin & Chakraborty, 2019).

In our study, the GC3 content varied greatly, falling between 27.96% and 88.98%, and in terms of compositional constraints, it was vital. In a previous study, freshwater turbellarians, cestodes, and trematodes exhibited substantial variation in their GC3 composition. This variation led to the speculation that these organisms could be divided into two distinct groups based on the range of their GC3 content (Khandia, Saeed, et al., 2022; Lamolle et al., 2019).

RSCU analysis revealed selective pressures influencing codon usage in blood group system genes

The overall RSCU value indicates the codon usage pattern for a gene by revealing overrepresented, underrepresented and randomly used codons. The average RSCU values for all genes associated with different blood group systems were calculated and categorized into four categories: (a) overrepresented codons (RSCU > 1.6), (b) underrepresented codons (RSCU< 0.6), (c) more often used codon (RSCU > 1) and (d) rarely used codons (RSCU < 1). Almost half of the codons in group 1 (26/59) and group 2 (31/59) were frequently used. The average of all CDSs RSCU values (Table 3) in this study shows that 11 C-ending codons (TTC, ATC, AGC, CCC, ACC, GCC, TAC, CAC, AAC, GAC, GGC) and six G-ending codons (CTG, GTG, CAG, AAG, GAG, AGG) were preferred over AT-ending codons. The matrix plot for RSCU values, the underrepresented codons (0.6), and the overrepresented codons (>1.6) are shown in Figure 1.

Additionally, a comparative analysis of thirteen neurodegenerative disorder-associated blood groups (group 1) with the rest of the blood group systems (group 2) showed that the C-ending and G-ending codons are preferred over T-ending and A-ending codons, respectively. Moreover, the G-ending codons such as CTG for leucine and GTG for valine were preferred in both groups and were found the most overrepresented codon. Likewise, the C-ending codons such as ATC for isoleucine, and GCC for alanine were preferred in both groups and were found the most overrepresented codon (Table 3). However, AGC that codes for serine was preferred in group 1 while TCC was preferred for group 2 genes. Similarly, CCT that codes for proline was preferred in group 1 while CCC was preferred for group 2 genes. In the present study, CTG (Lecuine), ATC (Isoleucine), GTG (Valine), AGC (Serine), and GCC (Alanine) were found over-represented codons (RSCU > 1.6) in group 1 genes where ATC for Isoleucine, and AGC for serine, explicitly found differed in synonymous codon choice than group 2 (Table 3). Our result is in concordance with a similar result found in a previous study on genes involved in the central nervous system (CNS) (Uddin & Chakraborty, 2019).

Additionally, two codons, CAA for glutamine were found underrepresented in group 1 only, whereas ACG for threonine was found underrepresented in group 2 genes only.

The findings reported here are comparable to a study of the genes associated with chronic periodontitis (Khandia, Pandey, et al., 2022), where elevated T→C transition rates might be responsible for its predominance by producing an abundance of CTG codons. Also, PCA analysis (Figure 11)

Table 3. RSCU values of Group 1, Group 2, and the average of all 263 transcripts' codons in 44 envisaged blood group system genes.

Amino acid	Codons	Group 1 RSCU	Group 2 RSCU	Overall RSCU
Phenylalanine (F)	TTT	0.766	0.821	0.807
	TTC	1.234	1.179	1.193
Leucine (L)	TTA	0.311	0.530	0.476
	TTG	0.660	0.850	0.804
	CTT	0.729	0.695	0.703
	CTC	1.251	1.247	1.248
	CTA	0.431	0.376	0.389
	CTG	2.617	2.302	2.379
Isoleucine (I)	ATT	0.690	1.001	0.925
	ATC	1.921	1.578	1.661
	ATA	0.388	0.422	0.413
Valine (V)	GTT	0.761	0.585	0.628
	GTC	0.904	1.083	1.039
	GTA	0.281	0.342	0.327
	GTG	2.054	1.989	2.005
Serine (S)	TCT	0.835	1.105	1.039
	TCC	1.414	1.324	1.346
	TCA	0.923	1.036	1.008
	TCG	0.332	0.309	0.314
	AGT	0.894	0.904	0.901
	AGC	1.602	1.322	1.391
Proline (P)	CCT	1.255	1.106	1.142
	CCC	1.215	1.213	1.213
	CCA	1.101	1.131	1.124
	CCG	0.428	0.551	0.521
Threonine (T)	ACT	0.825	0.888	0.873
	ACC	1.352	1.429	1.410
	ACA	1.201	1.204	1.203
	ACG	0.622	0.479	0.514
Alanine (A)	GCT	0.990	1.069	1.050
	GCC	1.829	1.636	1.683
	GCA	0.824	0.960	0.927
	GCG	0.358	0.335	0.340
Tyrosine (Y)	TAT	0.717	0.875	0.837
	TAC	1.283	1.125	1.163
Histidine (H)	CAT	0.771	0.773	0.772
	CAC	1.229	1.227	1.228
Glutamine (Q)	CAA	0.555	0.759	0.709
	CAG	1.445	1.241	1.291
Asparagine (N)	AAT	0.681	0.867	0.822
	AAC	1.288	1.133	1.171
Lysine (K)	AAA	0.773	0.754	0.759
	AAG	1.227	1.246	1.241
Aspartic acid (D)	GAT	0.693	0.880	0.835
	GAC	1.307	1.100	1.150
Glutamic acid (E)	GAA	0.778	0.772	0.774
	GAG	1.222	1.228	1.226
Cytosine (C)	TGT	0.817	0.809	0.811
	TGC	1.183	1.131	1.143
Arginine (R)	CGT	0.520	0.347	0.389
	CGC	1.190	1.048	1.082
	CGA	0.461	0.542	0.523
	CGG	1.064	1.026	1.035
	AGA	1.284	1.575	1.504
	AGG	1.480	1.463	1.467
Glycine (G)	GGT	0.842	0.891	0.879
	GGC	1.284	1.238	1.249
	GGA	0.909	0.903	0.904
	GGG	0.965	0.968	0.967

The values in red font represent the preference of the respective codon among all synonymous codons of an amino acid. The codons with values >1.6 are overrepresented and values < 0.6 are underrepresented and are highlighted with green and red, respectively.

implicated that CTG contributed to the maximum which is followed by CGC in the context of G/C ending codons. In Biplot analysis, the longer the line greater the variation. Consequently, selective pressures appear to be engaged in the overrepresentation of CTG codons, which substantially impacts overall codon usage.

Z-test implies a significant deviation in codon usage between group 1 and group 2 blood group system genes

Z-test is a key statistical approach to measure variance between a given population provided the sample size is large ($n \geq 30$). A paired z-test/related z-test is employed to compare two equivalent sets of results where they are linked (*Numeracy, Maths and Statistics - Academic Skills Kit*, n.d.). Z-test using RSCU values of group 1 and group 2 blood group systems revealed that 61.01% of codons have a significant amount of variation in codon usage between both the groups. Moreover, 25.42% codons (GCT, TTC, ATC, ACC, GCC, CAC, TAC, AAC, GAC, TGC, CTG, GTG, CAG, AAG, GAG) represented a highly significant ($p < 0.01$) variation, whereas 35.59% codons (TTT, TCT, AGT, CCT, ACT, TAT, AAT, GAT, TGT, CCA, ACA, GCA, AAA, CTC, GTC, TCC, AGC, CCC, GGC, AGG, GGG) represented a slight significant ($p < 0.05$) variation, implicating a substantial difference in codon usage by the genes of both the groups (Table 4).

Effects of mutational bias in CUB of blood group system genes

Like the CAI, the RCA is determined by taking the geometric mean of the RCAxyz term for each codon xyz in the entire sequence of interest. It is also suitable for algorithmic iterations that do not require a prior biological understanding to determine the reference set. However, unlike the CAI, the RCA index first determines the expected frequency of a codon based on its positional base frequencies for any given reference set. Following that, it assesses codon adaptation by measuring the deviation between the observed and expected codon frequencies. As a result, the RCA is likely to yield a more robust and precise estimate of gene expression, as it explicitly considers sequence composition when computing each RCAxyz term. Based on an earlier study, it is speculated that this enhanced accuracy is particularly evident in cases of high levels of mutational bias and/or reduced selection for translational efficiency, where the CAI might be duped by artifacts attributed to mutational bias in the reference set. Because of its increased accuracy in determining expression levels across various organisms (*Mycobacterium smegmatis* and *Escherichia coli*) and contexts, the RCA index is more appropriate for predicting gene expression (Fox & Erill, 2010).

In our study, RCA shows a significant ($r = 0.4532^{***}$, $\rho = 0.00017$) linear positive correlation with gene length for blood group systems genes associated with neurodegenerative disorders (Group 1) while a slight yet significant ($r = 0.1452^{*}$, $\rho = 0.040$) linear positive correlation with gene length for Group 2 blood group systems genes (Figure 2).

A dinucleotide study showed that CpG suppression influences codon usage

Dinucleotide biases have been shown to influence CUB. In this study, relative dinucleotide abundance analysis indicated that among all 44 blood group system genes, TpG (1.396)

Table 4. Z-test using RSCU values performed between Group 1 and Group 2 implicated 36 codons with highly significant variation.

Codons	Z	p (two tailed)	Significance
TTT	−2.1562	0.0310	*
CTT	−1.845	0.0650	NS
ATT	−1.9374	0.0526	NS
GTT	−1.2395	0.2151	NS
TCT	−2.2886	0.0221	*
AGT	−2.0156	0.0438	*
CCT	−2.3819	0.0172	*
ACT	−2.1012	0.0356	*
GCT	−2.8171	0.0048	**
TAT	−2.0126	0.0441	*
CAT	−1.6339	0.1022	NS
AAT	−2.0359	0.0417	*
GAT	−2.1229	0.0337	*
TGT	−2.0036	0.0451	*
CGT	−1.0007	0.3169	NS
GGT	−1.8999	0.0574	NS
TTA	−0.95169	0.3412	NS
CTA	−1.5385	0.1239	NS
ATA	−1.4255	0.154	NS
GTA	−1.2905	0.1968	NS
TCA	−1.6366	0.1017	NS
CCA	−2.5128	0.0119	*
ACA	−2.409	0.0159	*
GCA	−2.0667	0.0387	*
CAA	−1.5315	0.1256	NS
AAA	−1.9901	0.0465	*
GAA	−1.9186	0.0550	NS
CGA	−1.0453	0.2958	NS
AGA	−1.4666	0.1424	NS
GGA	−1.8619	0.0626	NS
TTC	−3.185	0.0014	**
CTC	−2.3636	0.0180	*
ATC	−2.7142	0.0066	**
GTC	−2.3221	0.0202	*
TCC	−2.1528	0.0313	*
AGC	−2.1119	0.0346	*
CCC	−2.0895	0.0366	*
ACC	−2.8288	0.0046	**
GCC	−2.9044	0.0036	**
CAC	−2.5968	0.0094	**
TAC	−2.7987	0.0051	**
AAC	−2.8783	0.0039	**
GAC	−2.8815	0.0039	**
TGC	−2.7019	0.0068	**
CGC	−1.1876	0.2349	NS
GGC	−2.1096	0.0348	*
CTG	−2.5924	0.0095	**
TTG	−1.893	0.0583	NS
GTG	−3.245	0.0011	**
TCG	−1.4992	0.1338	NS
CCG	−1.3671	0.1716	NS
ACG	−1.2982	0.1942	NS
GCG	−1.1612	0.2455	NS
CAG	−2.7882	0.0053	**
AAG	−3.2544	0.0011	**
GAG	−3.0418	0.0023	**
CGG	−1.2162	0.2239	NS
AGG	−2.2438	0.0248	*
GGG	−2.2798	0.0226	*

A significant variation among 15 codons usage is highlighted in brown representing the significance at $^{**}p < 0.01$ while the other 21 codons are shown with the significance at $^{*}p < 0.05$.
$^{**}p < 0.01$; $^{*}p < 0.05$; NS: not significant.

has the highest dinucleotide content, while CpG (0.494) has the lowest (Figure 3a). A comparative analysis of observed/expected frequencies of dinucleotide content is shown in Figure 3b. In vertebrates, genomes with low CpG content have high TpG plus CpA content to a nearly comparable degree. Strong evidence for the evolutionarily-driven conversion of methylated CpG to TpG plus CpA is speculated by this observation (Bird, 1980). Remarkably, we spotted a slight yet significant correlation ($r = 0.375^{*}$, $p = 0.011$) in the whole blood group system genes between the average spike in TpG and CpA dinucleotides and the loss of CpG dinucleotides (Figure 4). A past investigation findings ($r^2 = 0.77$; $p < 0.0001$) has also speculated that methylated cytosines deamination within CpG dinucleotides causes the loss of CpG dinucleotides and simultaneously spike in TpG/CpA dinucleotides (Upadhyay et al., 2013). Since cytosine in the presence of CpG is known to be methylated in eukaryotes and transforms into methylated cytosine, the underrepresentation of CpG makes sense (Salser, 1978). There is experimental evidence supporting the under-representation of CpG and TpA, dinucleotides in eukaryotes, and CpG in mammalian genomes (Alqahtani et al., 2022; Karlin & Mrázek, 1997; Kumar et al., 2021; Simmen, 2008). Moreover, we discovered underrepresented TpA and CpG dinucleotides in our research, which we anticipate to be a consequence of selectional factors (Khandia et al., 2024; Munjal et al., 2020). Also, TpA dinucleotide is adversely selected because it is an integral part of two of the three stop codons (TAG and TAA) and has significance in the mRNA destabilizing effect (Alqahtani et al., 2021).

Furthermore, in a past study of 35 CSFV genomes (Tao et al., 2009), the findings suggest that the composition of dinucleotides supports differential mutational pressure, which influences the disparity in the pattern of synonymous codon utilization between diverse CSFV ORFs.

Additionally, CpC and GpG overrepresentation were observed in our study in concordance with our previous results obtained in genes inducing autophagy and apoptosis with reference to neurodegenerative disorders (Khandia et al., 2024) and proapoptotic *Bim* gene isoforms (Khandia, Ali Khan, et al., 2022). The comparative analysis resembled the overall result except for the CD36 blood group system of group 1 with high ApA and low CpC/GpG dinucleotide contents (Table 5). In several ailments, including Alzheimer's disease, Rett syndrome, fragile X syndrome, Parkinson's disease, and amyotrophic lateral sclerosis, CpC has been attributed to non-CpG methylation (Jang et al., 2017).

Rare codon analysis indicated serine encoding codon TCG was the rarest while varied rare codon choice in both groups

Rare codons indicate such codons that appear in a gene or transcript occasionally. Rare codons may cause highly expressed genes to express at a reduced rate due to a decreased supply of the corresponding tRNAs (Frumkin et al., 2018). Rare codons have the potential to prevent translation initiation itself and to trigger ribosome stalling at the 5′ end (Liu et al., 2021). The 5′ end of the transcript contains an assembly of rare codons that assure appropriate protein folding and biological activity (Bentele et al., 2013; Rosano & Ceccarelli, 2009). Codons with occurrence frequencies of less than 0.5% were considered rare.

In this study, serine encoding codon TCG was the rarest in all genes. Moreover, valine encoding codon GTA was rare in

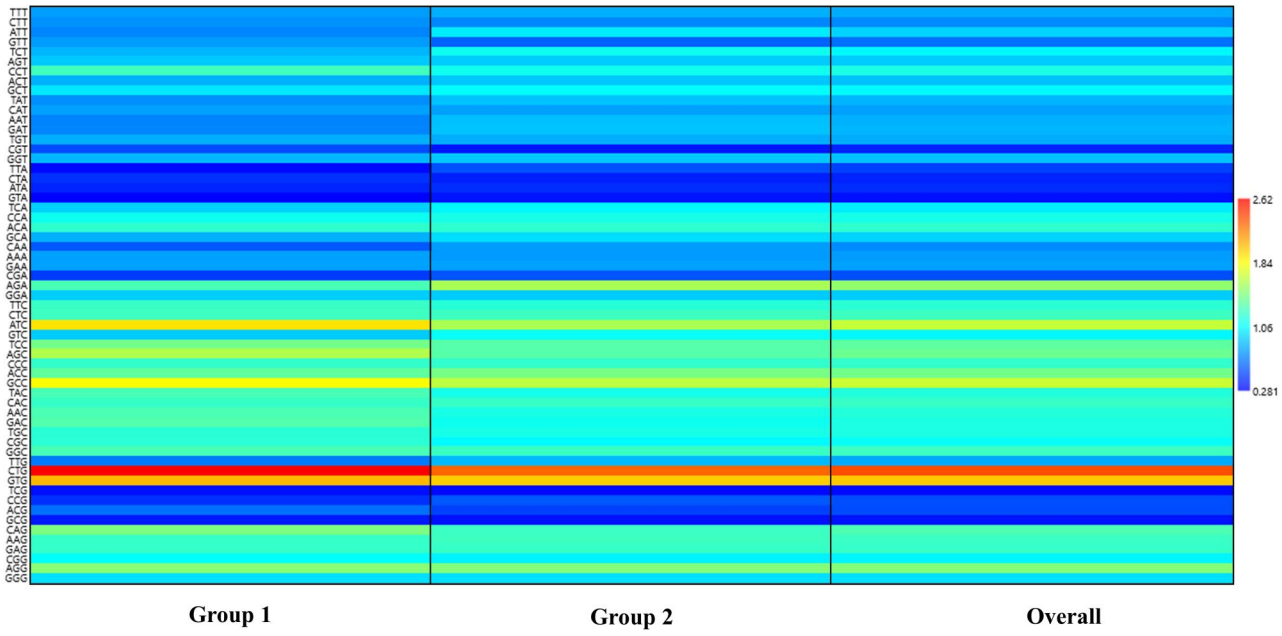


Figure 1. RSCU heatmap of Group 1, Group 2, and average of 263 transcripts of envisaged blood group system genes.

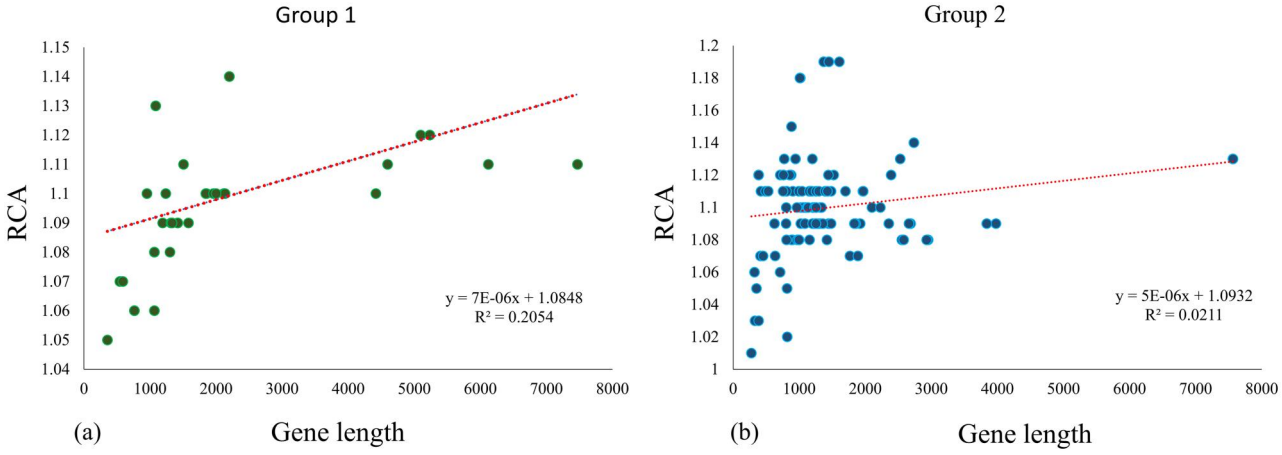


Figure 2. Distribution of RCA values versus transcripts length of (a) 64 transcripts for Group 1 and (b) 199 transcripts for Group 2 of envisaged blood group system genes.

group 1 genes while two codons for arginine CGA and CGT were found rare in group 2 genes (Figure 5). In our previous study, CGT codon was found rare in all four genes (APP, CCND1, CCNE2 and PTPA) associated with neurodegeneration and cancer (Khandia et al., 2023). In the present study, CGT (0.52%) frequency was observed closer to rare codons in overall blood group system genes. Additionally, In human cytomegalovirus (HCMV) genes, the six codons, GCG (Ala), CCG (Pro), CGT (Arg), CGC (Arg), TCG (Ser) and ACG (Thr) were observed rare (Kanduc, 2017). Knowledge of rare codon frequencies could be beneficial for modifying multiple genes.

Compositional attributes and codon preference indicated relevance in codon usage

The average ENc and ENc′ values of all CDSs of blood group systems are 50.26 (range 33.57-61) and 50.28 (range 39.83-54.14) respectively, indicating a lower codon usage choice (Table 2). Extreme bias is shown by a value of 20, whereas no bias is indicated by a value of 61, which implies no propensity in available codons utilization (Khandia et al., 2019; Uddin, 2024). There is a difference observed between ENc and ENc′ average values between group 1 and 2. In group 1 genes, ENc and ENc′ average values are 46.55 ± 5.29 and 50.66 ± 2.49, respectively, whereas in group 2 genes ENc and ENc′ values are 51.45 ± 6.88 and 50.16 ± 3.22 (Supplementary Tables 2 and 3). A high ENc value found in a previous study on CNS genes (ENc 40.53) (Uddin & Chakraborty, 2019), and genes associated with PD (ENc 42.74) (Uddin, 2024), implicated low CUB. It is anticipated that a low CUB will encourage the effective use of different codons, which will accelerate translation. Optimal replication in various cell types with varying codon choices might be linked with lower CUB. In vertebrates with distinct cell types and varying codon choices, the presence of low CUB may aid in effective replication (Deb et al., 2020).

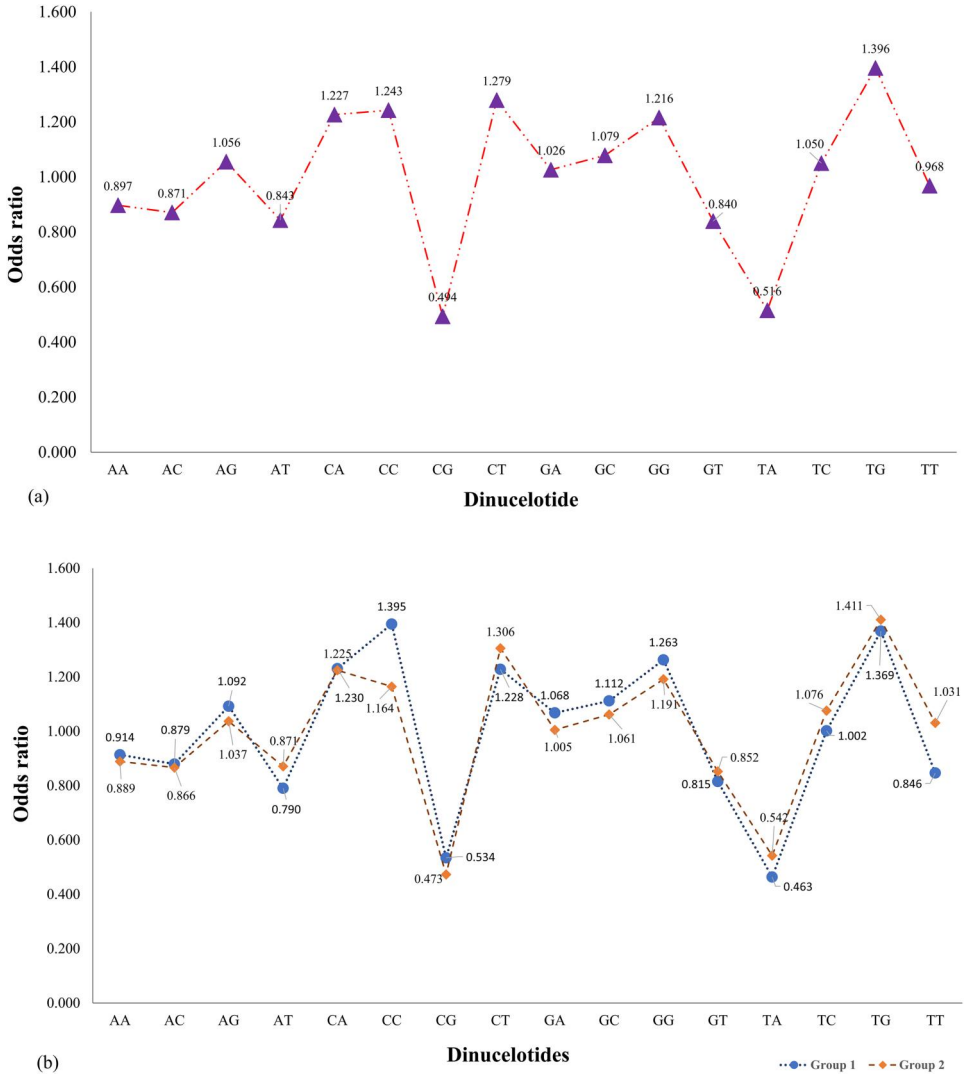


Figure 3. (a) Overall odds ratio (observed/expected frequencies) of all 263 transcripts of blood group system genes. The figure represents the highest dinucleotide content for TpG (1.396) while the lowest for CpG (0.494). (b) Comparison of odds ratio (observed/expected frequencies) between Group 1 and Group 2 blood group system genes.

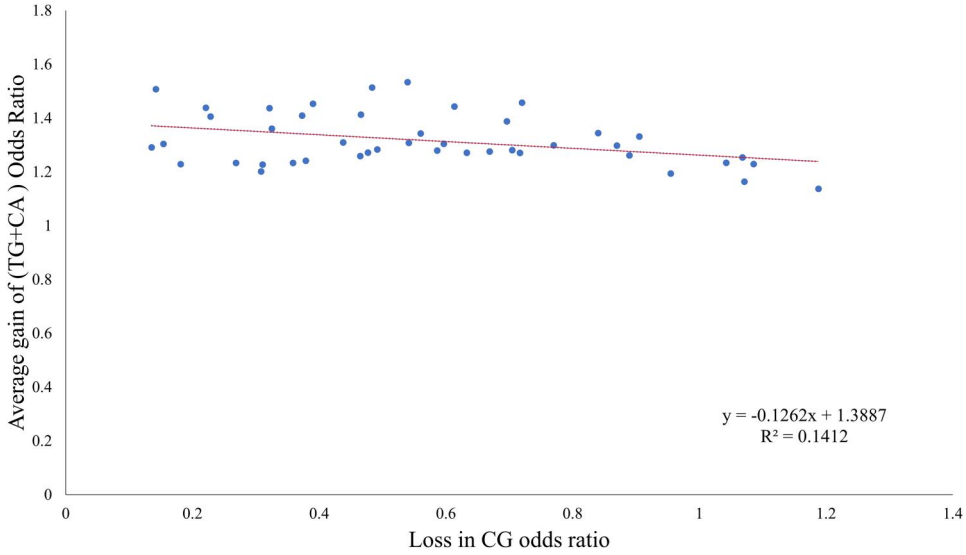


Figure 4. Correlation between loss of CpG dinucleotide versus average gain of (TpG + CpA) dinucleotides linked to 5-methylcytosine deamination. The plot represents a strong negative correlation between the odds ratio of CpG dinucleotide and the average odds ratio of TpG + CpA dinucleotides.

Table 5. Average dinucleotide content values of each 44 blood group systems.

		Dinucleotide															
	Blood groups	AA	AC	AG	AT	CA	CC	CG	CT	GA	GC	GG	GT	TA	TC	TG	TT
Group 1	ABCC1	0.731	0.850	1.094	0.724	1.257	1.353	0.705	1.275	1.145	1.136	1.356	0.822	0.266	1.250	1.305	0.731
	ABO	0.636	1.002	0.950	0.496	1.119	1.565	1.043	1.066	0.983	1.373	1.613	0.986	0.347	0.853	1.349	0.620
	Cartwright	0.348	0.896	0.948	0.387	0.983	1.967	0.955	1.281	0.991	1.480	1.927	0.837	0.257	0.844	1.405	0.494
	Chido	0.651	0.829	1.265	0.586	1.194	1.558	0.586	1.326	1.175	1.260	1.465	0.781	0.310	1.018	1.364	0.632
	CD36	1.798	0.813	1.017	1.334	1.243	0.633	0.154	0.980	1.023	0.584	0.729	0.929	0.898	0.979	1.365	1.521
	CTL2	0.744	0.844	0.759	0.914	1.058	1.349	0.633	1.530	0.945	1.191	1.208	0.739	0.512	1.186	1.485	0.904
	Kell	0.758	0.901	1.260	0.658	1.328	1.584	0.438	1.399	1.164	1.222	1.157	0.604	0.328	1.042	1.292	0.866
	Kx	0.816	0.720	1.127	0.804	1.163	1.139	0.492	1.547	0.852	1.175	0.804	0.996	0.636	1.307	1.403	1.019
	Knops	1.185	0.832	1.057	1.030	1.273	1.301	0.325	1.102	1.065	0.882	1.061	0.884	0.581	0.986	1.448	0.990
	Kanno	0.694	1.009	1.114	0.778	1.493	1.198	0.484	1.030	1.009	1.261	1.997	0.862	0.399	0.736	1.535	0.399
	Lewis	0.413	1.194	0.929	0.442	1.312	2.094	0.870	1.430	0.914	1.504	1.312	0.664	0.339	0.914	1.283	0.383
	John Milton Hagen	0.615	1.060	1.116	0.492	1.356	1.804	0.770	1.295	0.975	1.406	1.400	0.738	0.328	0.955	1.242	0.448
	Xg	1.300	1.118	1.042	0.870	1.386	1.606	0.669	0.822	1.070	1.013	1.147	0.793	0.573	0.746	1.166	0.679
Group 2	Augustine	0.636	0.686	1.003	0.762	1.358	1.446	0.390	1.607	0.766	1.412	1.085	0.764	0.327	1.258	1.548	0.951
	CD59	1.409	0.953	0.912	0.746	1.202	0.788	0.373	1.492	0.829	1.036	0.870	1.036	0.580	1.078	1.617	1.078
	EMM	0.864	0.889	1.083	0.742	1.130	0.989	0.542	1.335	1.051	1.096	1.182	0.958	0.528	1.021	1.486	1.105
	Er	0.349	0.848	1.098	0.464	1.284	1.698	0.905	1.409	0.868	1.770	1.550	0.743	0.256	0.980	1.379	0.399
	Gebrich	0.462	0.894	1.187	1.002	1.495	1.572	0.539	1.064	1.156	1.356	1.141	0.786	0.432	0.848	1.572	0.493
	Gill	0.342	0.805	0.674	0.684	1.167	1.860	0.613	1.468	0.764	1.458	1.579	0.784	0.231	0.986	1.720	0.865
	Indian	1.159	1.213	1.313	0.969	1.688	1.146	0.322	0.984	1.333	0.832	1.009	0.654	0.473	0.948	1.186	0.770
	LAN	0.475	0.888	0.999	0.634	1.227	1.064	0.696	1.461	0.969	1.428	1.649	0.846	0.325	1.067	1.549	0.722
	MAM	0.456	0.847	0.749	0.717	1.141	1.369	0.717	1.792	0.815	1.271	1.303	0.782	0.358	1.532	1.401	0.749
	MNS	1.575	1.136	0.997	1.340	1.297	0.707	0.311	1.061	1.211	0.450	0.568	0.804	0.965	1.082	1.157	1.340
	PEL	1.380	0.803	1.154	1.090	1.112	0.728	0.379	1.048	1.220	0.785	0.977	0.896	0.712	0.952	1.371	1.395
	RHAG	0.937	0.846	0.898	1.198	1.250	0.846	0.221	1.263	1.002	0.963	1.146	0.781	0.690	0.924	1.627	1.406
	SID	0.851	0.934	1.294	0.673	1.263	1.579	0.477	1.095	1.208	0.906	1.242	0.937	0.429	0.995	1.280	0.837
	Colton	0.451	0.989	0.834	0.655	1.316	1.594	0.840	1.378	0.890	1.390	1.625	0.754	0.260	1.156	1.372	0.494
	P1PK	0.513	1.071	0.799	0.437	1.237	2.081	1.086	1.372	0.875	1.553	1.312	0.679	0.196	1.071	1.221	0.498
	Rh	0.684	0.689	0.898	0.812	1.143	1.018	0.466	1.519	0.820	1.199	1.410	1.028	0.437	1.239	1.683	0.955
	H	0.379	0.948	0.802	0.540	1.123	1.838	0.890	1.517	0.846	1.502	1.371	0.744	0.321	1.079	1.400	0.700
	Raph	0.430	0.958	1.263	0.441	1.339	1.056	0.720	1.467	1.071	1.746	1.708	0.743	0.253	0.822	1.576	0.407
	Lutheran	0.448	0.909	1.209	0.379	1.157	1.961	1.071	1.226	1.088	1.731	1.679	0.631	0.252	0.815	1.170	0.272
	Kidd	0.923	0.828	0.819	0.985	1.318	1.332	0.229	1.455	0.757	0.978	1.266	0.807	0.559	1.194	1.494	1.055
	Landsteiner-Wiener	0.615	0.778	0.908	0.403	0.970	1.700	1.188	1.359	0.799	1.823	1.489	0.778	0.321	0.915	1.304	0.649
	Duffy	0.395	0.685	0.854	0.564	1.070	1.681	0.142	1.987	0.632	1.455	1.486	0.838	0.385	1.059	1.945	0.822
	Diego	0.433	0.772	1.094	0.673	1.217	1.626	0.597	1.539	1.012	1.363	1.457	0.708	0.310	1.217	1.392	0.591
	Dombrock	1.198	1.073	1.215	1.045	1.446	1.102	0.136	1.367	1.062	0.864	0.763	0.559	0.825	1.011	1.136	1.198
	Junior	1.325	0.840	1.120	1.218	1.247	0.690	0.269	1.170	1.102	0.682	0.894	0.824	0.830	1.163	1.220	1.405
	I	1.432	1.064	0.988	0.918	1.087	1.054	0.308	1.359	1.087	0.792	0.905	0.733	0.796	0.898	1.316	1.260
	FORS	0.409	1.020	1.070	0.618	1.418	1.738	0.560	1.293	0.890	1.228	1.648	0.779	0.399	1.023	1.268	0.639
	Ok	0.597	1.081	1.214	0.426	1.373	1.535	1.068	1.185	1.202	1.498	1.527	0.718	0.146	1.047	1.135	0.250
	Cromer	1.438	1.001	1.092	0.939	1.288	1.164	0.465	1.001	0.951	0.817	0.824	1.008	0.783	0.936	1.229	1.064
	Scianna	1.009	0.785	1.278	0.706	1.099	1.413	0.359	1.424	1.256	0.942	1.200	0.796	0.404	1.155	1.368	0.807
	Globoside	1.266	0.884	0.998	1.331	1.185	0.902	0.181	1.063	1.135	0.461	0.803	0.856	0.894	1.085	1.272	1.683

The red-marked values show less than 0.78 (underrepresented) and the green-marked values show greater than 1.25 (overrepresented).

The ENc value correlated positively with %A, %T, %AT, %A3, %T3 and %AT3, while negatively correlated with %G, %C, %GC, %C3, %G3, %GC3, CAI, RCA and PC1 indicating nucleotide compositions and codon preference as a key contributor in driving the CUB among CDSs of envisaged blood group systems (Table 6). In a past study, the correlation between ENC and %A, %T, %A3 and %T3 in human autophagy genes was found to be significantly positive whereas a negative correlation was reported between ENC and %C, %G, %GC, %C3, %G3. The study revealed that except AGG and TTG codons, all the G- and C- ending codons were positively correlated with %GC3 while A- and T- ending codons were negatively correlated with %GC3. These findings displayed the impact of G/C-ending codons in human autophagy genes (Jamil et al., 2022). Our findings is in concordance with the result of a past study of genes involved in neurodegeneration with iron accumulation, revealing a high significant negative correlation of ENc with %GC and %GC3 (Alqahtani et al., 2021). If the content of these two nucleotides (G and C) is raised, it would likely result in increasing bias.

Moreover, a correlation between %GC3 and all the 59 codons, except AGG and TTG all G- and C-ending codons were positively correlated with %GC3, whereas A- and T-ending codons were negatively correlated with %GC3 (Figure 6). In contrast, the correlation between ENc and all the 59 codons revealed that all the G/C-ending codons showed negative correlation with ENc except TTG and AGG, whereas all the A/T-ending codons (except CGT) showed positive correlation with ENc (Figure 7). This result is in concordance with the correlation between ENc and nucleotide composition. A similar result was reported in our previous study where a positive correlation of A/T-ending codons with ENc while negative correlation with G/C-ending codons (except TTG-Leu) in genes involved in neurodegeneration with iron accumulation indicated that an elevated GC content will suppress the usage of leucine coding TTG codon (Alqahtani et al., 2021).

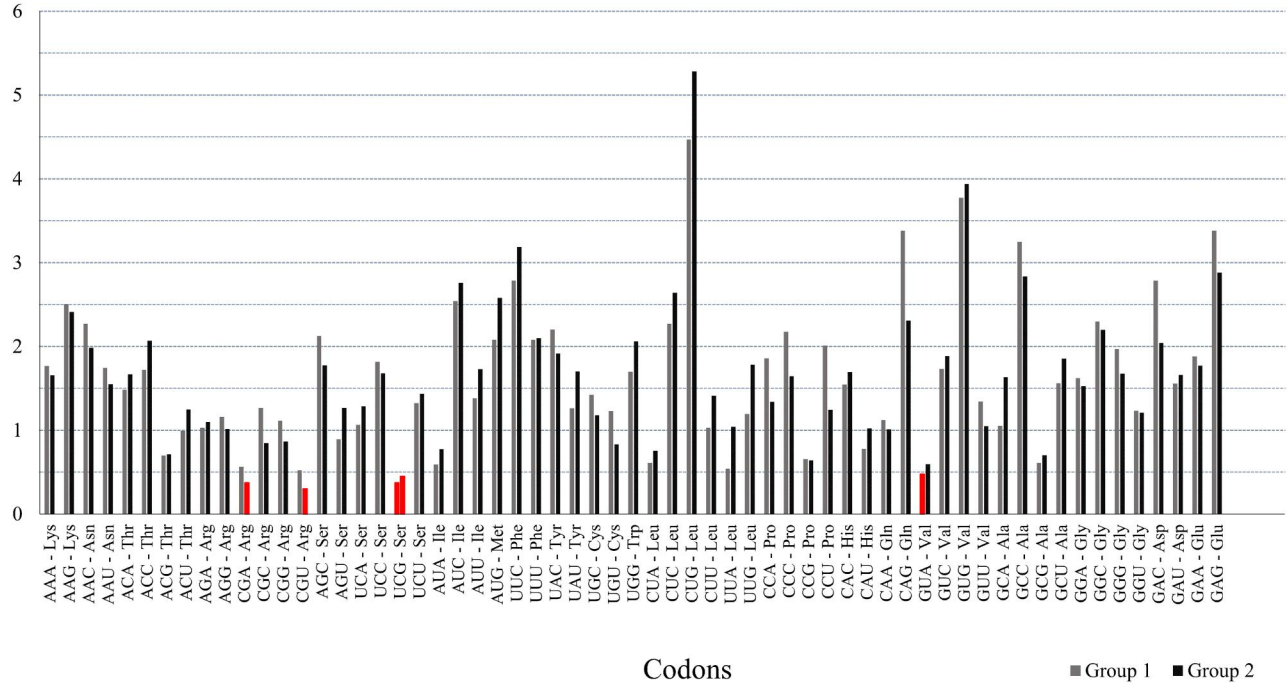


Figure 5. Rare codon analysis between Group 1 and Group 2 genes. The grey color bars represents the value for overall transcripts of Group 1 and the black color bars represent the Group 2. The codon(s) with a frequency less than 5 in 1000 are depicted in red color bars and are rare.

Furthermore, codons ending with G or C at the synonymous position were preferred by the genes as indicated by the heat map (Figure 1).

Relevance of evolutionary constraints and the association between overall nucleotide composition and third codon position

The nucleotide composition at the third codon position and overall nucleotide composition were correlated in order to determine if natural selection and/or mutation pressure had a greater impact on the course of evolution. A substantially significant ($p < 0.001$) positive correlation was identified between the third codon position and the homogenous nucleotide composition in the genes of the entire blood group system, while other nucleotide comparisons, as indicated in Table 7, showed a substantial ($p < 0.001$) negative correlation. A complex correlation between the compositions of nucleotides was observed. In particular, there is a substantial positive correlation between %A, %T, %AT, and %A3, %T3, %AT3, likewise, %C, %G, %GC, and %C3, %G3, %GC3, while a substantial negative correlation between %A, %T, %AT, and %C3, %G3, %GC3, likewise, %C, %G, %GC, and %A3, %T3, %AT3. These findings imply that the use of synonymous codons might be influenced by the nucleotide constraints. Similar results were found in a previous study of mitochondrial cytochrome oxidase genes in fishes, birds, and mammals (Uddin et al., 2019), suggesting role of both natural selection and mutational pressure in compositional constraints. Our result is also in concordance with a similar result found in a previous study on CNS genes (Uddin & Chakraborty, 2019). In addition, the regression analyses showed slope values ranging between 0.4689 and 0.5637, implying that in all blood group system genes, codon usage has been influenced by both natural selection and mutational pressure (Figure 12).

CAI value close to 1 indicates selection has been effective in molding the pattern of codon usage

CAI value can be used to evaluate gene expression purely based on its codon composition. The average CAI value for all 263 CDSs of 44 blood group systems was 0.761 ± 0.04 (range 0.62 to 0.87), indicating a considerable opportunity for gene expression (Table 2). The correlation of CAI with nucleotide composition reveals that significant positive correlations were found between CAI and %G, %C, %GC, %C3, %G3, %GC3, RCA, PC1, and PC2 whereas significant negative correlations were found between CAI and %A, %T, %A3, %T3, %AT, %AT3, and ENc (Table 6). In particular, the genes that express themselves at higher levels exhibit greater CUB and GC-rich content. Our result follows the work of Khandia et al., (2022) (Khandia, Pandey, et al., 2022), who found a significant negative correlation with ENc, and a positive correlation with %GC, and %GC3 in the transcripts associated with chronic periodontitis. Moreover, these genes exhibited preferences for codons ending with G or C at synonymous sites. We conclude that nucleotide composition and the degree of gene expression have significance in all blood group system genes.

Role of background nucleotide compositions in measuring CUB

The CDC analysis fully addresses background nucleotide compositions (BNCs) across diverse sequences and at the three codon positions by accounting for both GC and purine

Table 6. Correlation analysis between ENC, CAI, RCA, PC1, PC2, and nucleotide contents (%A, %C, %T, %G, %G+C, %A+T, %A3, %C3, %T3, %G3,%GC3, %AT3, and P2).

	CAI	RCA	PC 1	PC 2	%A	%C	%T	%G	%G + C	%A + T	%A3	%C3	%T3	%G3	%GC3	%AT3	P2
ENC	−0.858***	−0.247***	−0.833***	−0.073	0.624***	−0.787***	0.775***	−0.686***	−0.784***	0.784***	0.750***	−0.819***	0.811***	−0.668***	−0.805***	0.805***	0.187**
CAI	–	0.400***	0.898***	0.181**	−0.649***	0.842***	−0.817***	0.699***	0.822***	−0.822***	−0.830***	0.892***	−0.832***	0.686***	0.857***	−0.857***	0.004
RCA	–	–	0.277***	0.214***	−0.313***	0.314***	−0.105	0.123*	0.238***	−0.238***	−0.249***	0.273***	−0.246***	0.194**	0.255***	−0.255***	0.016
PC 1	–	–	–	–	−0.877***	0.930***	−0.819***	0.865***	0.954***	−0.954***	−0.937***	0.948***	−0.964***	0.874***	0.981***	−0.981***	0.002
PC 2	–	–	0.0008	–	0.185**	0.049	−0.115	−0.144*	−0.044	0.044	0.091	0.046	0.021	−0.176**	−0.057	0.057	0.800***

Significance values at $p < 0.001$ are marked with three stars (***), $p < 0.01$ are marked with two stars (**), $p < 0.05$ are marked with one star (*) and unmarked values are not significant at any of these p values.
***$p < 0.001$, **$p < 0.05$, **$p < 0.01$, NS: not significant.

contents. Thus, CDC analysis effectively determines comparative magnitudes and codon usage patterns for genes with diverse nucleotide compositions. In the present study, the average CDC value for all 263 CDSs of 44 blood group system genes was 0.117 ± 0.03 (range 0.07 to 0.22), indicating a lower amount of CUB (Table 2). A previous study analyzing CDC outcomes revealed that CDC exhibits substantially reduced biases compared to scaled ENc, particularly when the purine content has varied positional compositions. Additionally, for the majority of parameter combinations examined by Zhang et al. (2012), CDC consistently surpassed ENc when they defined inconsistent positional BNCs for both GC and purine contents. However, the CDC displayed identical values only when the positional composition sets were consistent because ENc overlooked BNCs (Zhang et al., 2012). These speculations agree well with those of Novembre (2002) (Novembre, 2002).

Neutrality plot analysis indicated supremacy of selection pressure in shaping the CUB

Neutrality analysis was employed to determine the degree of evolutionary constraints in shaping the CUB of different blood group system genes (Kumar et al., 2016). A significant positive correlation coefficient between the GC12 and GC3 indices suggests that codon utilization trends are influenced by mutational pressure (U. Kumar et al., 2021). Our findings revealed a significant positive correlation ($r = 0.657^{***}$, $p < 0.00001$; GC12 versus GC3), implying mutational pressures in nearly each codon position. Besides that, a regression line slope of less than 0.5 indicates selection pressure (N. Sueoka, 1988); the regression slope was 0.269 (Figure 8a), indicating neutrality of 26.92% vis-à-vis, the mutational pressure of 26.92%, and selection force of 73.08%, illustrating how natural selection has contributed to codon usage among all blood group system genes. Also, a comparative neutrality plot for Group 1 and Group 2 is shown in Figure 8b.

A disproportionate distribution of bases in parity analysis indicated the identical role of evolutionary constraints in shaping the CUB

The parity plot analysis was employed to attain greater insight into how mutational pressure and natural selection affect the codon usage of the blood group system's genes. The four-fold degenerate codon can be utilized to determine the neutral mutation rate. AT or GC should be used identically among a gene's degenerate codon groups provided mutation pressure is solely responsible for establishing CUB (Zhang et al., 2007). Nonetheless, for codon determination, natural selection would not necessarily contribute to an equal distribution of G and C or A and T (Noboru Sueoka & Kawanishi, 2000). The majority of the spots among 44 blood group systems on a PR2 bias plot were found in the bottom right corner (+X,-Y quadrant), indicating that T and G were utilized more frequently in the third codon position than A and C in different blood group system genes (Sophiarani & Chakraborty, 2022) (Figure 9).

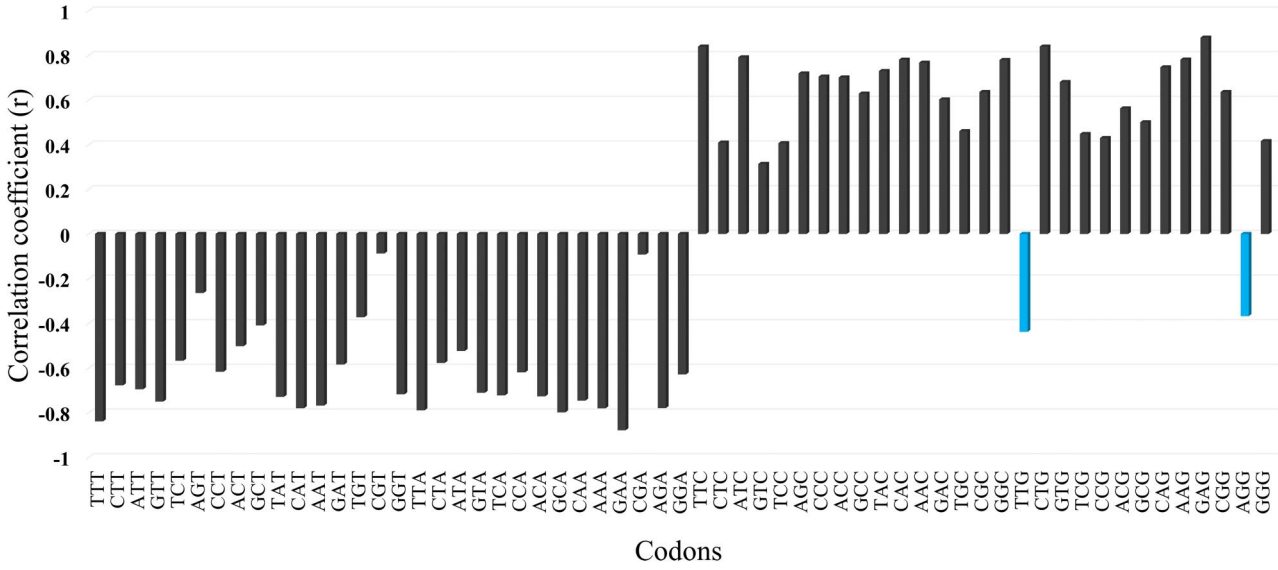


Figure 6. Correlation between %GC3 and all 59 codons. The graph depicts the correlation between %GC3 (r values) and all 59 codons of 263 transcripts of blood group system genes, where the G/C ending codons show a positive correlation while the A/T ending codons show a negative correlation with %GC3. The blue color bars represent the negative correlation of TTG and AGG codons with %GC3.

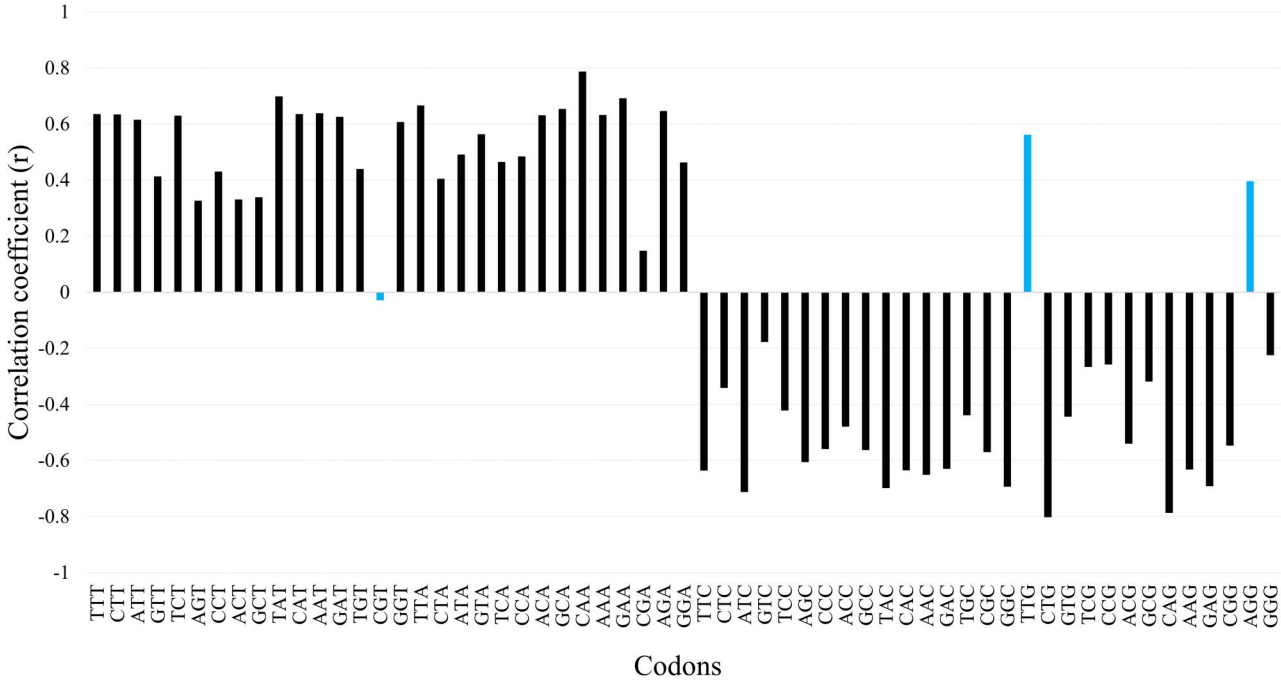


Figure 7. Correlation between ENc and all 59 codons. The graph depicts the correlation between ENc (r values) and all 59 codons of 263 transcripts of blood group system genes, where the A/T ending codons shows a positive correlation while the G/C ending codons show a negative correlation with ENc. The blue color bars represent the positive correlation of TTG and AGG codons with ENc.

Table 7. Correlation analysis between overall nucleotide contents and third codon position (%A, %C, %T, %G, %G + C, %A + T, %A3, %C3, %T3, %G3, %GC3 and %AT3).

	%A3	%C3	%T3	%G3	%GC3	%AT3
%A	0.891***	−0.805***	0.840***	−0.865***	−0.892***	0.892***
%C	−0.818***	0.912***	−0.914***	0.738***	0.894***	−0.894***
%T	0.643***	−0.755***	0.818***	−0.642***	−0.755***	0.755***
%G	−0.815***	0.732***	−0.841***	0.877***	0.854***	−0.854***
%GC	−0.866***	0.878***	−0.933***	0.851***	0.928***	−0.928***
%AT	0.866***	−0.878***	0.933***	−0.851***	−0.928***	0.928***

Significance values at $p < 0.001$ are marked with three stars (***), $p < 0.01$ are marked with two stars (**), $p < 0.05$ are marked with one star (*) and unmarked values are not significant at any of these p values.
***$p < 0.001$, *$p < 0.05$, **$p < 0.01$, NS: not significant.

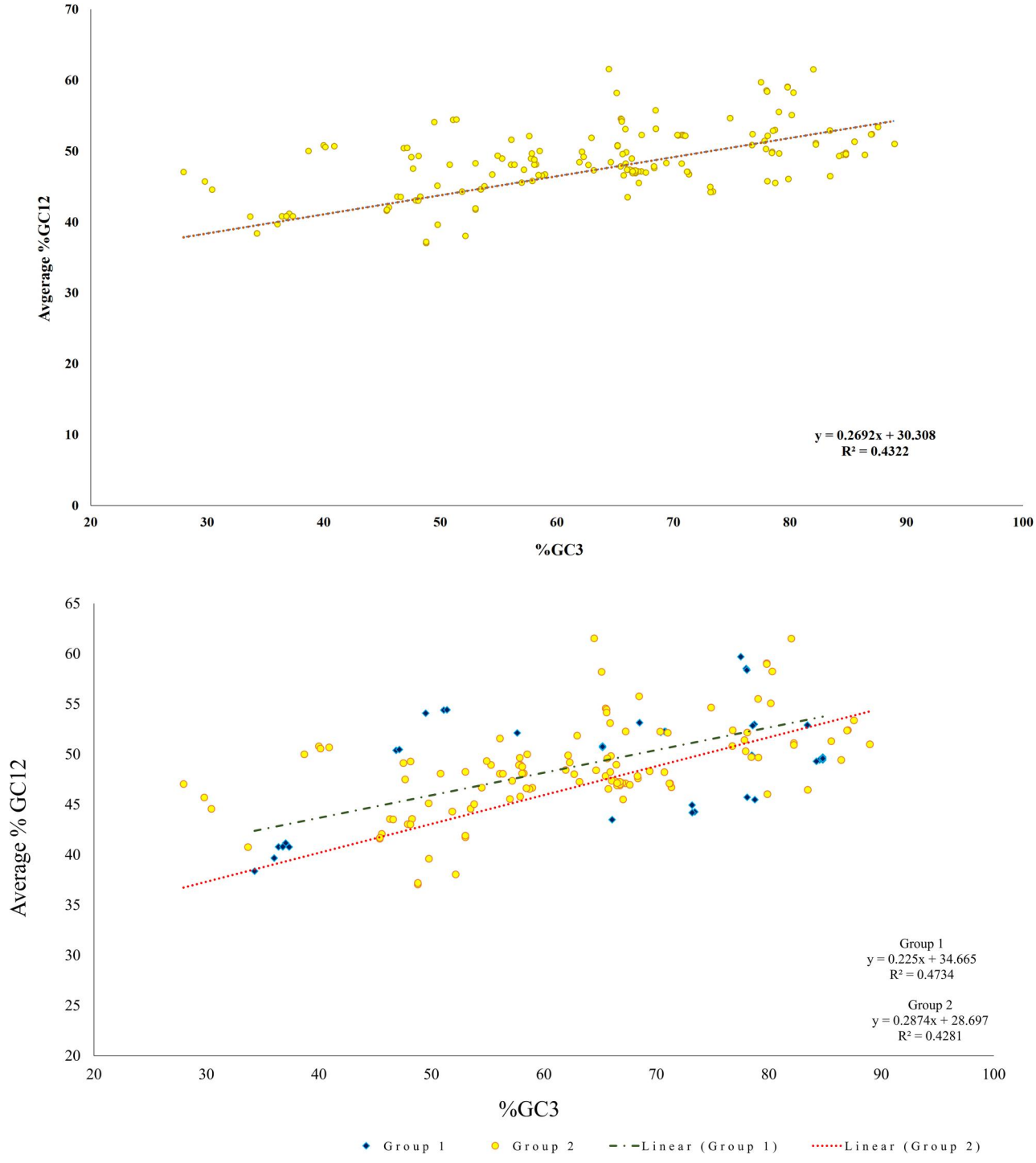


Figure 8. (a) Neutrality plot analysis (%GC12 vs %GC3) for 263 transcripts of envisaged blood group system genes. %GC12 stands for the average values of GC contents at the first and second positions of the codons (GC1 and GC2), likewise, GC3 refers to the GC contents at the third codon position. The yellow circle points and the corresponding trend line is the linear regression of %GC12 against %GC3 (r = 0.657, $p < 0.00001$), (b) Neutrality plot analysis (%GC12 vs %GC3) for grouped genes. %GC12 stands for the average values of GC contents at the first and second positions of the codons (GC1 and GC2), likewise, GC3 refers to the GC contents at the third codon position. The blue diamond points and the corresponding trendline is the linear regression of %GC12 against %GC3 (r = 0.688, $p < 0.00001$), Relative neutrality is 22.5% for Group 1, while the yellow circle points, and the corresponding red trendline is the linear regression of %GC12 against %GC3 (r = 0.654, $p < 0.00001$), Relative neutrality is 28.74% for Group 2.

Our results suggest T preference over A as the average position of $y = 0.343 \pm 0.148$ (AT bias), likewise G preference over C as the average position of $x = 0.566 \pm 0.143$ (GC bias). As the value exceeds 0.5, purine is preferred over pyrimidine. The values were not located in the center of the parity plot. An unequal distribution could mean that both mutational and selection factors are at work in shaping the CUB of blood group systems (U. Kumar et al., 2021). Additionally, the outcomes of the compositional and dinucleotide analyses were found to be consistent with the parity analysis, where a high TpG odds ratio suggested that T/G was preferred over A/C (Table 2).

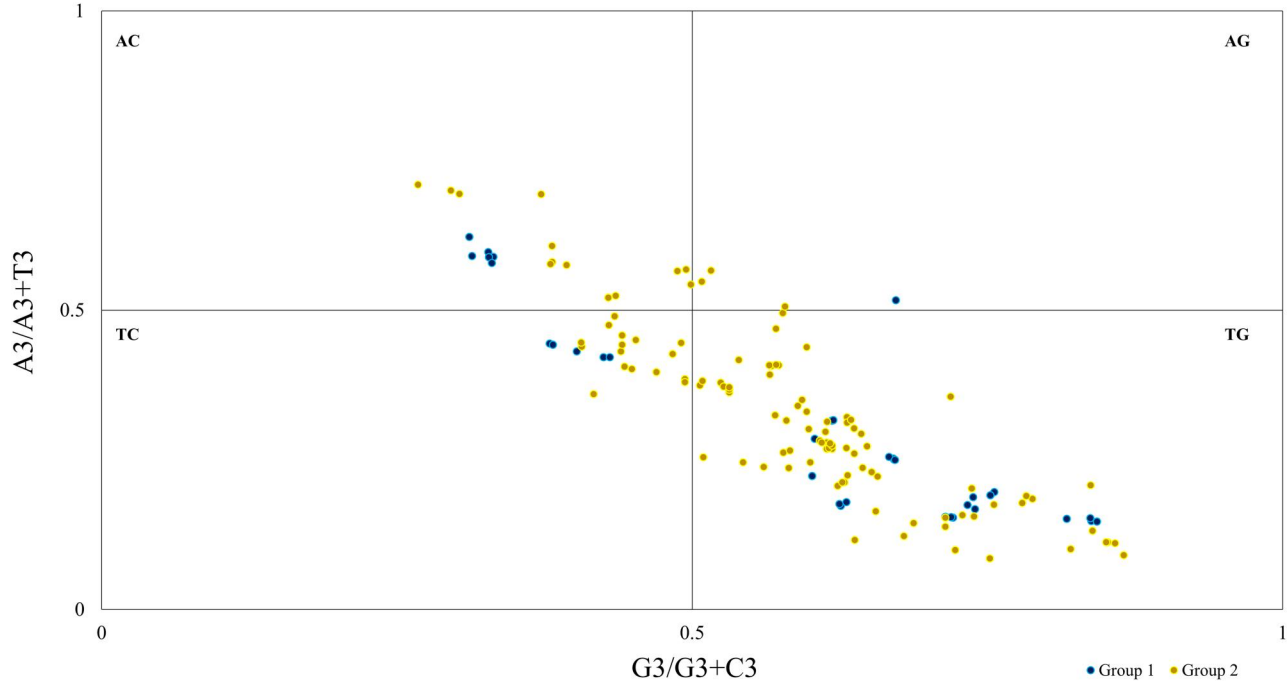


Figure 9. Parity analysis indicating nucleotide T preference over A, and G preference over C. The blue spots depict the 64 transcripts for Group 1 and the yellow spots depict 199 transcripts for Group 2 of envisaged blood group system genes, where both the plots suggest the preference of T and G over A and C, respectively.

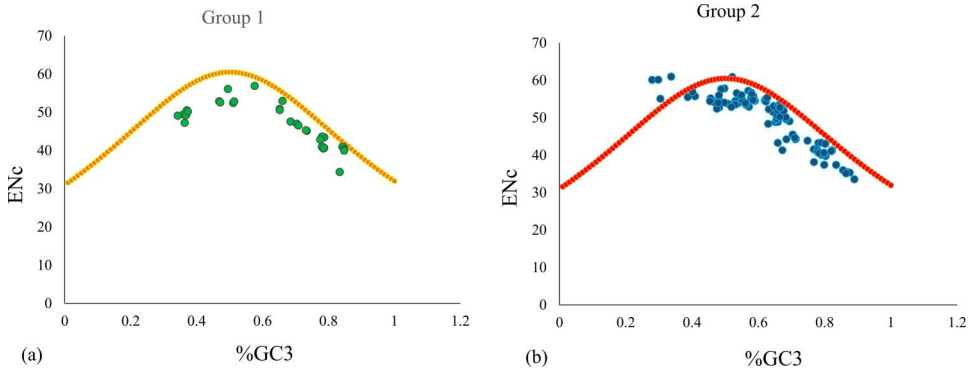


Figure 10. ENc versus %GC3 plot (selection curve). (a) The values of ENc and %GC3 of 64 transcripts for Group 1 (green spots) and (b) values of 199 transcripts for Group 2 (blue spots) of blood group system genes. ENc denotes the effective number of codons and GC3s denotes the GC content in the third synonymous codon position. The curve in red represents the expected curve when the codon usage was only determined by the GC3 composition.

ENc-GC3 curve analysis deciphered the major role of natural selection in configuring the CUB

The ENc-GC3s plot is often employed to decide whether a gene's codon utilization is solely the result of mutational or selectional pressure alone. If the measured values plunge on the predicted curve, mutational pressure is present; if such points plunge beneath the predicted curve, selectional pressure is present (Khandia et al., 2019). Figure 10a shows that selectional pressure determines codon usage in CDSs of group 1 while a few CDSs of group 2 lie on the predicted curve, indicating both selection and mutation pressure influence the CUB (Figure 10b). According to reports, one important component affecting the bias in codon usage across genomes is GC composition (Wan et al., 2004). Table 6 shows a substantial ($p < 0.001$) correlation between ENc and G3 (r=-0.668), and ENc and C3 (r = -0.819), suggesting that ENc influences the CUB. These findings shed light on the variance in codon usage for GC constraints and expand our knowledge of the molecular makeup of the blood group system's genes (Coghlan & Wolfe, 2000).

P2 analysis indicated the dominance of translational selection over mutation pressure in group 1 genes implicating substantial gene expression

With P2 values greater than 0.5 ($P2 = 0.512 \pm 0.05$) for each CDSs in the envisaged group 1 (Supplementary Table 2), it was evident that translational selection prevailed over mutation pressure in codon usage (Bahiri-Elitzur & Tuller, 2021). However, in group 2, a P2 value less than 0.5 ($P2 = 0.498 \pm 0.06$) indicated mutational force over translational selection in CUB determination (Chakraborty et al., 2019) (Supplementary Table 3). Similar results to group 1 were found in a previous study of genes related to dementia, where a high P2 value (1.01) indicated a strong translation efficacy pointing to selectional pressure (Alqahtani et al., 2022).

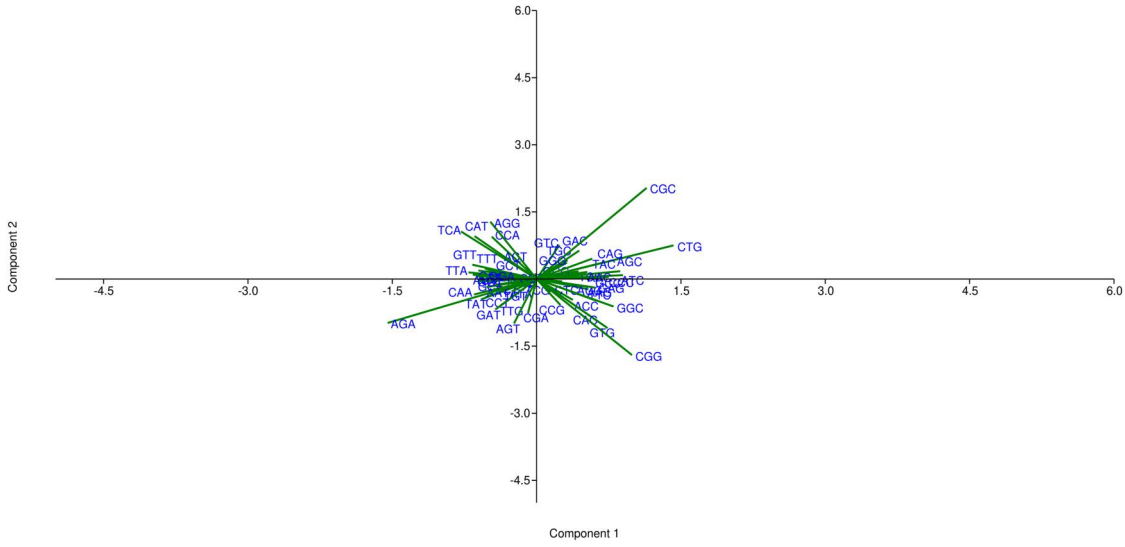


Figure 11. PCA analysis. Biplot analysis showed that CTG (Leu) and CGC (Arg) along the first axis, AGA (Arg) along the third axis, and CGG (Arg) along the fourth axis impacted CUB greatly. The longer the line, the greater the variation.

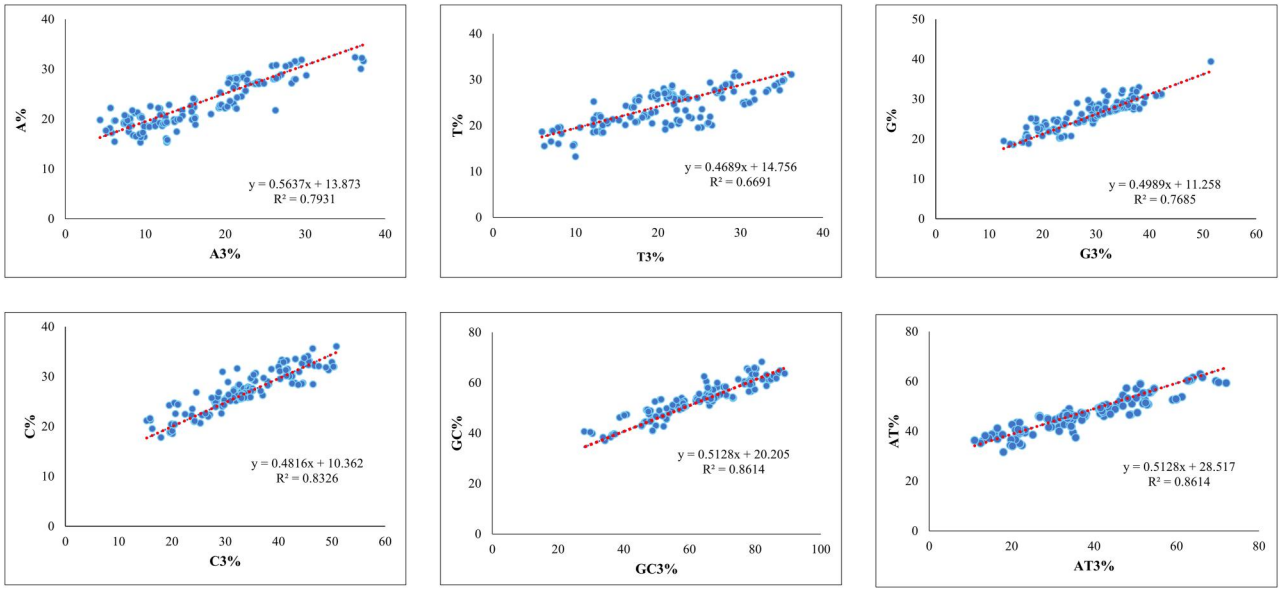


Figure 12. Regression analysis. The graph depicts the regression analysis between overall nucleotide bases (A%, T%, G%, C%, GC% and AT%) and the overall third position of nucleotide bases (A3%, T3%, G3%, C3%, GC3% and AT3%).

A negative correlation between P2 and ENc (−0.653**), speculated an inverse relationship between translational selection and CUB (Deb et al., 2020). Also, CDSs with low ENc values have high translational selection during evolution. However, in the present study, the correlation between P2 and ENc is highly positive ($r = 0.187^{**}$, $p = 0.002$) indicating a direct relationship of translational selection with CUB (Table 6).

Principal component analysis (PCA)

To emphasize the corresponding distribution of the 59 variations in codon usage and estimate the distributions of each vector, PCA was performed. The PCA findings revealed that the first major axis accounts for 46.31% of all variants, while the second, third and fourth axes account for 8.05%, 6.40% and 4.75% of all variations, respectively. The analysis inferred that the first four axes accounted for 65.52% of the total, whereas, the first axis plays a significant role in all 59 mutations, specifically a tendency for codon use bias. Moreover, Biplot analysis implicated the maximum contribution of AGA followed by CTG and CGC, the longer the line more the variation (Khandia, Pandey, et al., 2022) (Figure 11). Biplot analysis showed that CTG (Leu) and CGC (Arg) along the first axis, AGA (Arg) along the third axis, and CGG (Arg) along the fourth axis impacted CUB greatly. Yang et al, (2010) speculated CGC (Arg), AGC (Ser), and GGC (Gly) codons' significance in influencing the CUB in Alzheimer's disease and other neurological ailments (Yang et al., 2010). Subsequently, the correlation between the first axis, ENC and compositional indices was investigated. We found a noteworthy positive correlation

between PC1 and %C, %G, %GC, %C3, %G3, %GC3, RCA and CAI, while a significant negative correlation with %A, %T, %AT, %A3, %T3, %AT3 and ENC (Table 6). A substantial correlation was observed between nucleotide compositions and synonymous codon usage, including axis 1, indicating that most of the CUB among CDSs of different blood group system genes were closely associated with base composition.

Furthermore, we assessed how PC1 and PC2 correlated with other indices (Table 6). The findings showed a positive correlation of PC2 with compositional indices (A%, CAI, RCA), and a negative correlation with (G%, G3%), implicating the impact of composition on the codon utilization.

Conclusion

The blood group system is composed of antigens that are controlled by one or more closely related loci, and it is highly polymorphic and genetically distinct. Genetic variations in ABO have been connected to the etiology of numerous systemic illnesses, including neurological ailments. Our study extends a certain understanding of molecular signatures linked to a group of genes that are substantially connected to the blood group system genes. Although blood groups are widely reported to have an impact on diseases, the relationship between many conditions—especially RhD status and blood types has not been thoroughly investigated. The blood group data may help in prediction models if clinicians are aware of the disease extent and vulnerability among other populations.

In our study, the comprehensive nucleotide composition analysis discovered GC-rich content in the blood group system genes, whereas two blood groups Knops and CD36 of group 1 indicated AT richness. An abundant GC3 content co-translates in particular spatial locations and possibly plays a role in the architecture of the genome. Additionally, RSCU analysis implicated G/C-ending codon preference while comparative analysis revealed distinct codon (AGC-Serine, CCT-Proline in group 1 and TCC-Serine, CCC-Proline in group 2) preference. Likewise, two codons (CAA-Glutamine in group 1 and ACG-Threonine in group 2) were found underrepresented. Moreover, leucine-coding CTG codon was the most overrepresented and was preferred in all blood group systems indicating selectional pressure substantially impacts overall codon usage. This is also supported by biplot analysis where CTG (Leucine) contributed maximum along the first axis. Z-test analysis showed a substantial difference in codon usage by the genes of both groups. A significant correlation between the average spike in (TpG + CpA) dinucleotides and the loss of CpG dinucleotides provided strong evidence for evolutionarily driven conversion of methylated CpG to (TpG + CpA). Additionally, CpC and GpG overrepresentation observed in the present study is in concordance with our previous results concerning neurodegenerative disorders where CpC has been attributed to non-CpG methylation and linked to several neurodegenerative ailments. However, the underrepresentation of TpA and CpG dinucleotides while the overrepresentation of CpC dinucleotides indicates a role of high-order selection force in governing codon usage. The selection curve also supports this. In contrast, parity and neutrality plot analyses suggested the role of other evolutionary constraints in addition to selectional pressure in shaping codon usage patterns and gene expression. However, the outcomes of the compositional and dinucleotide analyses were found to be consistent with the parity analysis, where a high TpG odds ratio suggested that T/G was preferred over A/C. Serine encoding codon TCG was the rarest in all genes while comparative rare codon analysis showed varied rare codon choices (GTA-Valine in group 1 and CGA, CGT-Arginine in group 2) in both groups. The rare codons have the potential to govern translation initiation and thus, the knowledge of rare codon frequency could be beneficial for modifying multiple genes and their expression level. RCA and CAI showed substantial gene expression in both groups. Also, a significant negative correlation between ENC-CAI and ENC-%GC3, infer that nucleotide composition and degree of gene expression have significance in all blood group system genes. Our results show new and intriguing connections between codon usage patterns and disease incidence, whilst strengthening previously reported associations with additional evidence. In order to evaluate possible causal relationships and biological interactions, this novel set of associations needs to be verified in further cohorts and investigated utilizing a mechanistic approach.

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Disclosure statement

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